

Week 2 Module:

Date: 17/06/2026

1. Fraud Transaction Detection (Linear Search)

Problem Statement

A bank stores transaction IDs in the order they occur. Since the transactions are not sorted, the security team needs to check if a suspicious transaction ID exists.

Write a program using **Linear Search**.

Input Format

- First line contains integer N
- Second line contains N transaction IDs
- Third line contains suspicious transaction ID

Output Format

Print:

Fraud Transaction Found

or

Transaction Not Found

Example

Input

```
5
1201 1305 1450 1408 1502
1450
```

Output

```
Fraud Transaction Found
```

2. Server Log Search (Binary Search)

Problem Statement

A cloud company stores server log IDs in sorted order. Engineers need to quickly verify if a log ID exists.

Write a program using **Binary Search**.

Input Format

- First line contains N
- Second line contains sorted log IDs
- Third line contains target log ID

Output Format

Print:

Log Found

or

Log Not Found

Constraints

- $1 \leq N \leq 10^5$

Example

Input

```
7
1001 1005 1010 1015 1020 1030 1040
1015
```

Output

```
Log Found
```

3. Sort Employee Salaries (Bubble Sort)

Problem Statement

A company wants to sort employee salaries in ascending order for payroll analysis.

Write a program using **Bubble Sort**.

Input Format

- First line contains N
- Second line contains salaries

Output Format

Print sorted salaries.

Example

Input

5
45000 32000 55000 28000 60000

Output

28000 32000 45000 55000 60000

4. Sort Website Response Time (Insertion Sort)

Problem Statement

A web monitoring system records website response times. New data comes continuously and must be inserted into the correct position.

Write a program using **Insertion Sort**.

Input Format

- First line contains N
- Second line contains response times

Output Format

Print sorted response times.

Example

Input

5
250 120 320 180 100

Output

100 120 180 250 320

5. Prioritize Bug Reports (Selection Sort)

Problem Statement

A software company assigns priority scores to bug reports. To resolve bugs efficiently, they want to sort priorities in ascending order.

Write a program using **Selection Sort**.

Input Format

- First line contains N
- Second line contains priority scores

Output Format

Print sorted priority scores.

Example

Input

5
9 2 8 1 5

Output

1 2 5 8 9